

Rubik Cube – Game Play Documentation

1. Introduction

Rubik Cube is a Unity puzzle game template that simulates the classic cube-solving challenge. Players can solve cubes ranging from **2×2 to 9×9**, with smooth rotation controls along the **X, Y, and Z axes**.

The project is lightweight, customizable, and extendable, making it a strong foundation for **puzzle games, educational tools, or creative cube-based mechanics**.

Core Objective

Solve the Rubik Cube by aligning all faces by color using layer rotations along X, Y, and Z axes.

2. Core Features

-  Supports cube variations from 2×2 to 9×9.
-  Smooth layer rotation along X, Y, Z axes.
-  Camera orbit to view the cube from any angle.
-  Orientation gizmo shows cube orientation in the top-right corner.
-  Customizable cube colors, materials, rotation speed, and sizes.
-  Lightweight and extendable for custom game mechanics or educational purposes.

3. Gameplay Overview

Game Flow

- **Load Scene:** Open `RubikCubeScene.unity` in the Scenes folder.
- **Interact with Cube:** Rotate layers using X, Y, Z icons to align cube faces by color.
- **Camera:** Orbit freely to inspect the cube from any perspective.
- **Completion:** Once all faces match by color, the puzzle is solved.



4. Scenes

- `RubikCubeScene.unity` → Main playable scene, includes cube setup and UI.



5. Folder Structure

All game files are organized under the following directory path:

```
Assets/RubikCube/  
├─ Documentation/  
├─ Materials/  
├─ Models/  
├─ Prefabs/  
├─ Scenes/  
├─ Scripts/  
└─ Textures/
```



Folder Descriptions



Documentation/ → Project documentation and reference materials.



Materials/ → Cube materials and color settings.



Models/ → 3D cube models and related assets.



Prefabs/ → Reusable cube prefabs for different sizes.



Scenes/ → Unity scene files (`RubikCubeScene.unity`).



Scripts/ → Scripts controlling cube logic, rotation, and game mechanics.



Textures/ → Texture assets used for cube faces and UI elements.



6. Dependencies & Third-Party Assets

- No third-party assets are required.
- All assets (models, materials, scripts, textures) are original and license-compliant.

7. Integration & Customization Notes

- **Cube Materials:** Change colors in [Materials/](#).
- **Prefabs:** Edit or create new cube prefabs in [Prefabs/](#).
- **Scripts:** Customize cube logic and controls in [Scripts/](#).
- **UI:** Modify or extend axis icons and cube controls as needed.

8. Summary

Rubik Cube offers a mentally engaging puzzle experience that combines logic, spatial reasoning, and precision.

Its customizable cube sizes, smooth rotation, and intuitive UI make it suitable for casual players, developers, and educational projects alike.

Optimized for performance and flexibility, Rubik Cube is a timeless puzzle game template that challenges your mind while allowing full creative customization.

 **Play. Rotate. Solve. Repeat.**